

OLADOSU SAMUEL

Acoustic-Engineering with Digital Signal Processing (DSP) System Controllers

Technical notes on DSP-based system controllers for hi-resolution audio, room correction, and multi-zone acoustic engineering.

SC-2 & SC-3 System Controllers

The **SC-2** and **SC-3** are professional Digital Signal Processing (DSP) system controllers developed by Wisdom Audio for high-performance home theatre, cinema and multimedia-room audio installations. They serve as the central processing unit of an audio system by managing signal routing, crossover processing, equalization, delay, room correction, and speaker optimization before audio reaches the power amplifiers.

Unlike a conventional audio processor, a system controller ensures that every loudspeaker receives a signal specifically optimized for its frequency range and acoustic environment.

System Role in the Audio Signal Chain

A typical professional signal chain is:

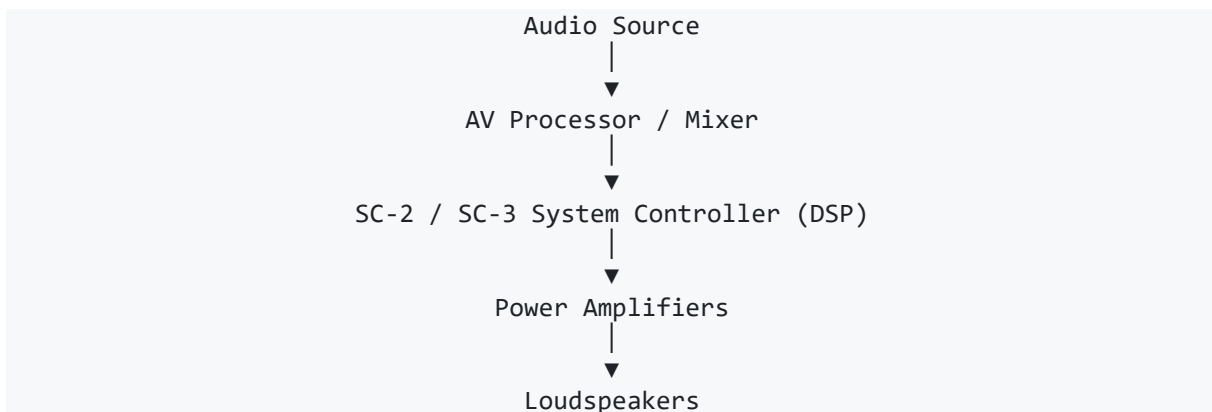


Figure 1 Audio Signal Chain

The system controller sits between the source processor and the power amplifiers, acting as the "brain" of the audio system.

Key Specifications

- Currently, there are two models SC-2 and SC-3
- These models are identical in performance and specifications but the only difference is the number of inputs and outputs.
- An SC-2 has an 8-channel input and 16 channel outputs.
- A SC-3 is a bigger unit of SC-2 that has 16-channel input and 32-channel outputs.

SC-2

- 8-channel input

- 16-channel output
- 24-bit / 96 kHz DSP processing
- Balanced XLR connections
- 2RU rack size

SC-3

- 16-channel input
- 32-channel output
- 24-bit / 96 kHz DSP processing
- Balanced XLR connections
- 3RU rack size

Unique Design Features

Advanced Room Correction: Dirac Live room calibration corrects for acoustic deficiencies introduced by the listening environment, improving clarity, bass definition, and tonal accuracy. The algorithm optimises both the time and frequency domain response of the audio system to extract maximum performance from the speakers. The SC supports bi- and tri-amplification, where dedicated amplifier channels drive discrete frequency bands — subwoofer, mid-bass woofers, and planar magnetic panels — independently through an active crossover, reducing distortion across the full audible range. The SC-2 accommodates up to 8 bi-amplifiable input channels and 16 Dirac-corrected output channels, each with independent user EQ and delay.

Multiple Setup Memories: The SC supports up to four complete, independent calibration presets, recallable on demand. Typical use cases include: - Variable room conditions (e.g., drapes open vs. closed) - Different target curves (e.g., music vs. gaming) - Multiple seating arrangements.

Bi/Tri-amplification: Active crossover architecture allows each frequency band to be driven independently by dedicated amplifier channels, reducing distortion across the full audible range.

System-Specific EQ: This feature employs equalization-based bass compensation to overcome the acoustic limitations imposed by their slim enclosures. By electronically correcting for the reduced low-frequency output inherent to small enclosure volumes (a technique widely adopted in subwoofer design). The system would achieve bass performance characteristic of significantly larger enclosures, without compromising the form factor.

Front Panel Indicators/Switch



Figure 2 SC 2 Front Panel

The SC-2 features a sculpted front panel that conceals two switches under the flexible ends of the middle section.

- LED 1: Power Switch, Standby Indicator and Input Clip Indicator.
 - Standby state: Amber LED is lit.
 - Input clipping state: Amber LED flashes rapidly for three seconds.
 - Power Switch State: Amber LED is OFF if it is the operate modes (fully on or fully off)
- LED 2: Setup Memory LED Indicators

Dirac® Live room correction enables addressing room- specific problems that would otherwise compromise performance. A single calibration of the system for the room and your preferred listening positions is all that is necessary. But the SC-2 provides storage of up to four complete and independent calibrations to address various situations.

The LEDs indicate which of the four-setup memories is active:

 - LED 1 on = Setup memory #1 enabled
 - LED 2 on = Setup memory #2 enabled
 - LED 3 on = Setup memory #3 enabled
 - LED 4 on = Setup memory #4 enabled
 - LED 1 blinking slowly = Dirac® bypassed temporarily
- Setup Calibration Selector Switch
 - Pressing the right side of the center section of the SC-2 front panel will activate a micro switch that cycles the unit through the available setup memories.
 - If only one memory has been set up (which is fairly typical), pressing this button will toggle Dirac on and off
 - If multiple setups have been stored, pressing this button will cycle through the available setups in order, ending in the same bypass mode for the room correction prior to starting the cycle over again:
 - Setup Memory 1
 - Setup Memory 2 (if available)
 - Setup Memory 3 (if available)
 - Setup Memory 4 (if available)
 - Bypass Dirac®

REAR PANEL

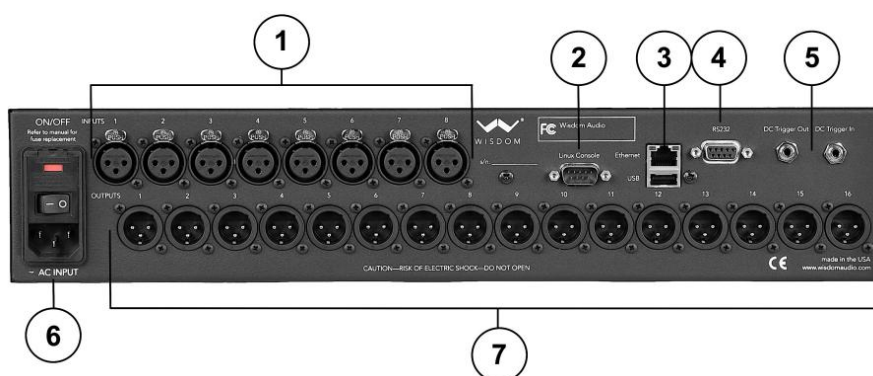


Figure 3 Rear Panel SC-2

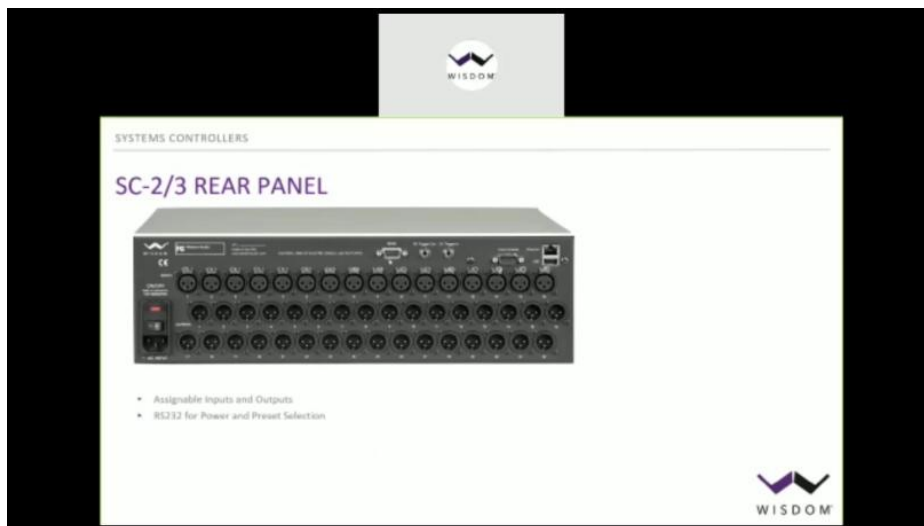


Figure 4 Rear Panel SC-3

1. 8 Channel Inputs
2. Linux Console
3. Ethernet & USB
4. RS232
5. DC Trigger In and Out
6. AC Mains Input & Switch
7. Outputs

Specification

- Internal Processing: 24-bit 96kHz
- Frequency Response: 20 Hz – 40k Hz ± 0.75 dB
- Signal to Noise ratio: -110 dB (ref 2.5V rms A-wtd.)
- Input impedance: $32k\Omega$ (balanced)
- Output impedance: 182Ω
- Maximum output voltage: 12.3V rms
- Maximum input voltage: 3.88V rms
- Power consumption in idle: 3W
- Maximum power consumption: 40W
- DC trigger input sensitivity: 4-24 Volts @ 5 mA
- DC trigger output signal: 12 Volts @ 150 mA
- Mains voltage ranges: 90-264 volts 47-63 Hz
- Weight: 30 lbs / 13.6 kg
- Dimensions HxWxD: 3.75" x 17" x 13.63" (8.57cm x 43.2cm x 34.7cm)
- USB mic needs to be connected to laptop and laptop is then connected to the SC wirelessly
- A SC controller can take care of 5.1 room
- System allows for multiple room to be set as 1 single platform
- Mini DSP, UMik, galaxy audio cricket tester and Dirac live for wisdom software
username and password are needed for a Dirac Live for wisdom calibration.

Configuring your SC-2 and Preparing for Calibration

Locating the SC on your network:

1. Device is configured via Ethernet/IP/HTML. Ensure Device is connected to LAN.

2. The device should always be discoverable at first startup by entering <http://SC.local/wizard.html#> into the address bar (This is the same log-on for both the SC-2 & SC-3, unless your unit has been upgraded to the new firmware then it is simply: <http://SC.local/wizard.html#>) of your web browser (Google Chrome works best).
3. If you cannot locate the device. The backdoor is to locate its IP address and access it directly. This can be done by IP search wizards such as Angry IP Scanner. Look for the MAC Vendor named “IEEE”. The MAC address should start off with Momentum Data System’s Mac Vendor ID: 70:B3:D5:E4:5X:XX
4. Once the IP address has been discovered use it to access the device, do not forget to add [/wizard.html#](#) to be taken to the Configuration Page. Example <http://192.168.1.10/wizard.html#>



Figure 5 Connecting to the SC-2 or SC-3

System Calibration

It is recommended to contact your dealer, or regional Wisdom Audio representative about purchasing a Factory Calibration service.